Machine Learning Analytics RSM8413 – Fall 2026

Group Assignment 1 Due date: 2026-11-13

Deliverables (to be submitted on Quercus):
1. Report including a detailed description of your findings. At most 10 pages + appendices.
2. All Python source code either in a Jupyter Notebook (*.ipynb) or a Python file (*.py). One file!
3. Contribution Statement: briefly describe each team member's contributions.
4. AI Appendix (if you used generative AI; see below).

Include an Executive Summary (at the beginning) describing your most salient findings. Explain all steps and results clearly and cogently, so that a reasonably intelligent though statistically naïve manager could understand it. You need to include all graphics in your report. Your narrative should be clear and concise, accompanied by supporting evidence in the form of graphics and tables. All tables and graphics should be well formatted (e.g., tables should not run over from one page to another).

The Case:
COVID-19 Behavior Data (COVID-19BehaviorData_CAN2022.csv)
Fresh from the Rotman MMA program, you are hired into your dream job: You join the Public Health Agency of Canada. You are tasked to study a data set collected by the Institute of Global Health Innovation Imperial College London. This data set includes information about symptoms, testing, isolation, social distancing, and other COVID-19-related behaviors for tens of thousands of individuals across 29 countries. The data come from interviews, with around 21,000 individuals interviewed each week.
Note: This is a very large data set, therefore, it has already been restricted to Canada only for data collected in 2022.

Tasks:

Perform data preparation on the data set.
Hint: As discussed in class, in multiple linear regression we typically create k–1 dummy variables for a categorical variable with k levels, to avoid perfect multicollinearity. However, for k-NN and tree-based models, this restriction is unnecessary – in fact, using all k dummy variables can work better. This way, each category gets its own column, and no single category is treated as the "baseline".

Perform EDA, especially with respect to the relationship between the predictor variables and the target, which you must choose! It must be a categorical variable. Clearly explain your choice and the underlying business objectives. Report only the most important results you uncover supported by suitable graphs.

What is a choice of k that balances between overfitting and ignoring the predictor information?

Tune the model: use k-fold cross-validation (e.g., 5-fold) with a grid search on the training data to tune k and at least one other hyperparameter (e.g., distance weighting of the votes, or the distance metric). Report the cross-validated performance of your candidate settings.
Carefully consider all points that we discussed in class with respect to k-NN.

Show the classification matrix for the test data that results from using the best model parameters.
Besides accuracy, report precision, recall (sensitivity), and the F1 score, and plot the ROC curve with its AUC for your final model on the test data. Which measure matters most for the business objective you chose, and why?
Discuss business implications: what do your results suggest for the Public Health Agency of Canada?

Team, Submission, and Academic Integrity:
This assignment is completed in your pre-assigned program team (the same team for the whole term). One team member submits all files for the team on Quercus by Friday, November 13, 2026, 23:59.
Your Jupyter Notebook must run from top to bottom on a fresh kernel (Kernel → Restart & Run All) without errors, loading the data files from the same folder via relative paths, so that every number, table and graph in your report can be reproduced.
Late submissions: 20% deduction if not received by the due date and time, and a further 10% deduction for each subsequent day.
All Quercus submissions go through the University's plagiarism detection tool, and your source code will be reviewed with Moss (Measure Of Software Similarity).

Use of Generative AI (GenAI):
Generative-AI tools (e.g., ChatGPT, Claude, Copilot) are permitted on this assignment, subject to the policy in the course outline. UofT-approved version of MS Copilot: https://m365.cloud.microsoft/chat
Disclosure. In an AI Appendix, state which tool(s) you used and for which parts of the work — for example, "Used ChatGPT to draft an initial version of the grid-search code; verified, tested, and edited by hand." Undisclosed use may be treated as an unauthorized aid.
Responsibility. You are accountable for everything you submit. Verify, test, and understand any code or analysis you turn in — "the AI gave me that answer" is not an acceptable defence.
Be exam-ready. The final exam (50%) is written without any AI. Use this assignment to practise the methods yourselves.